# Research Project R1: Bayesian modeling of transconjugacy efficacy under chemical stressors

Final Presentation

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Bayesian Statistics Course
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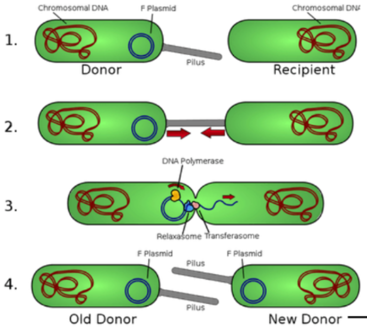
# Outline

# Overview

**Transconjugation** is the ability of a bacterium to receive genetic material, a **plasmid**, from another bacterium through **conjugation**.

This **horizontal gene transfer mechanism** contributes to the evolution and spread of diverse characteristics within bacterial populations, including traits such as **antibiotic resistance**.




**Bacterial Conjugation**

**Donor:** CL218 resistant to meropenem

**Recipient:** J53 not resistant to meropenem

→ Transconjugant

# Research Questions

**Goal**

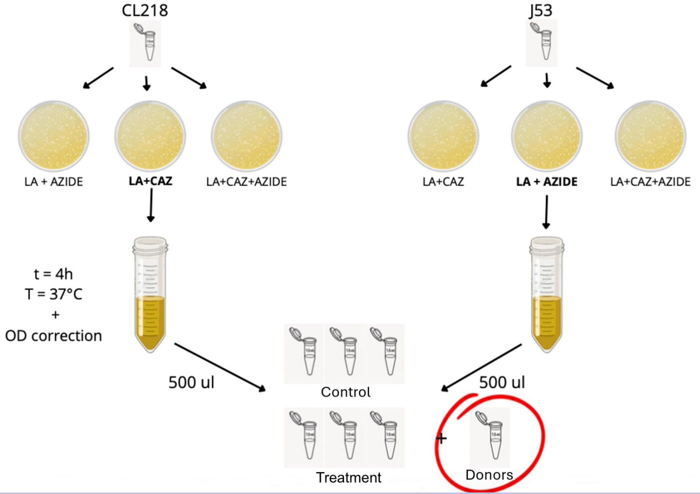
Modeling the frequency of transconjugation under different chemical stressors.

$\Rightarrow$

- How do **temperature**, **ibuprofen** and **DMSO** concentrations influence the **transconjugation rate per donor**?
- How much variability is introduced by **experimental replicates**, and what is the most appropriate way to model it?
- Can a **Bayesian model** quantify uncertainty in both biological effects and measurement noise?
- What is the **transconjugation response surface** across combinations of chemical stressors?

**Two strains of Escherichia coli bacteria** are used:

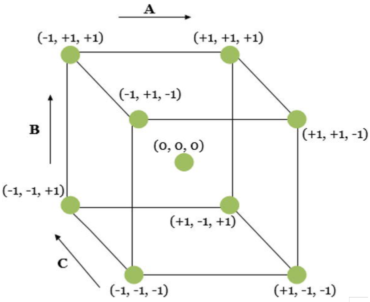
- **CL218**: Donor strain resistant to CAZ
- **J53**: Recipient strain resistant to AZIDE



# Design of Experiment

| Fattori | - | + |
|---|---|---|
| T [°C] | 28 | 40 |
| Ibuprofene [mg/l] | 6,25 | 50 |
| DMSO [%v/v] | 0 | 10 |

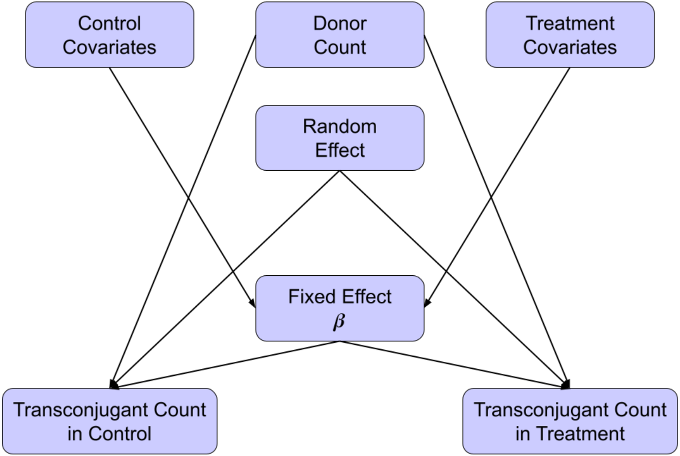
| # Esperimento | T [°C] | Ibuprofene [mg/l] | DMSO |
|---|---|---|---|
| 1 | - | - | - |
| 2 | + | - | - |
| 3 | - | + | - |
| 4 | + | + | - |
| 5 | - | + | + |
| 6 | - | - | + |
| 7 | + | - | + |
| 8 | + | + | + |




| # Esperimento | T [°C] | Ibuprofene [mg/l] | DMSO |
|---|---|---|---|
| 9 | - | 0 | + |
| 10 | + | 0 | + |

Different combinations of **temperature**, **ibuprofen concentration**, and **DMSO concentration** are tested to evaluate their effect on transconjugacy.

# Model Overview



# Generalized Linear Model

**Likelihood:**

$$T_{i,j,k,c} \mid \lambda_{i,j,c}, X_{i,c}, \rho_k \overset{\text{ind}}{\sim} \text{Poisson}(\rho_k\, \lambda_{i,j,c})$$
$$\lambda_{i,j,c} = \alpha_i\, \theta_{i,j,c}$$

- $T_{i,j,k,c}$ = transconjugant count
- $X_{i,c}$ = vector containing covariates (1, Temp, [IBU], [DMSO])
- $\rho_k = 10^{-\lceil \frac{k}{3} \rceil}$ dilution factor
- $\alpha_i$ = donor count (exposure) for experiment $i$
- $\theta_{i,j,c}$ = transconjugation rate per donor

Indexes:

- $i \in \{1, ..., 10\}$ = experiment
- $j \in \{A, B, C\}$ = experiment replica
- $k \in \{1, ..., 9\}$ = observations within each experiment replica
- $c \in \{0, 1\}$ = control (0) or treatment (1)

# Hierarchical Generalized Linear Model

**Linear predictor (log link):**

$$\log(\lambda_{i,j,c}) = \log(\alpha_i) + b_{i,j} + \boldsymbol{\beta}^\top F(\mathbf{X}_{i,c})$$

**Bayesian model for donors:**
**Likelihood:**

$$D_{i,k} \mid \alpha_i, \delta_k \overset{\text{ind}}{\sim} \text{Poisson}(\delta_k \, \alpha_i)$$

- $D_{i,k}$ = donor count measurement for experiment $i$
- $\delta_k = 10^{-5-\lceil \frac{k}{3} \rceil}$ dilution factor

Conditional independence between transconjugant and donor counts:

$$T_{i,j,k,c} \perp\!\!\!\perp D_{i,k} \mid \lambda_{i,j,c}, \rho_k$$

# Prior Specification

**Random effects:**

$$\begin{aligned} b_{i,j} \mid \sigma_b^2 &\overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_b^2) \\ \sigma_b^2 &\sim \text{Inverse-Gamma}(2, 1) \end{aligned}$$

**Fixed effects:**

$$\begin{aligned} \beta \mid \tau^2 &\sim \mathcal{N}(0,\ \tau^2 I_p) \\ \tau^2 &\sim \text{Inverse-Gamma}(2, 1) \end{aligned}$$

**Donor counts (exposure):**

$$\begin{aligned} \log(\alpha_i) \mid \mu_\alpha, \sigma_\alpha^2 &\overset{\text{iid}}{\sim} \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) \\ \mu_\alpha &\sim \mathcal{N}(0, 10^2) \\ \sigma_\alpha &\sim \text{half-}\mathcal{N}(0, 5^2) \end{aligned}$$

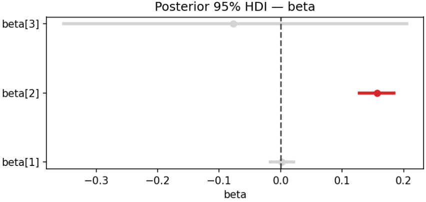
We assume a priori independence between the random effects $b$, the fixed effects $\beta$, and the donor exposures $\alpha$.

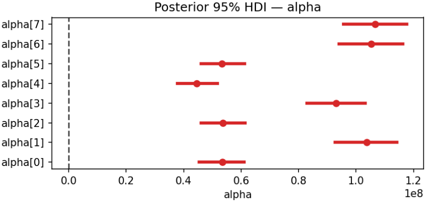
# Implementation

| Task | Configurations | N | H |
|---|---|---|---|
| Covariate selection ($\beta$) | Simple | 6 | 3 |
| | Lasso | 6 | 3 |
| | Horseshoe | 4 | 4 |
| | Regularized Horseshoe ($p_0 = 2$) | 4 | 4 |
| | Regularized Horseshoe ($p_0 = 1$) | 4 | 4 |
| | R2D2 | 4 | 4 |
| | Spike & Slab | 6 | 4.5 |
| Sensitivity Analysis | Inverse-Gamma | 12 | 6 |
| | Half-Normal | 6 | 3 |
| | Half-Student t | 4 | 2 |
| NB grid search | $6 \times 6$ grid search on $\phi_{D_{rate}}$ and $\phi_{T_{rate}}$ | 72 | 40 |
| | **Total** | **128** | **77.5** |

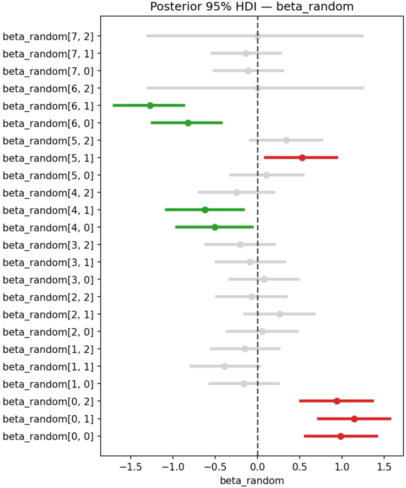
- N = number of configuration tested
- H = total hours spent for the corresponding row

## Simple Poisson Model:



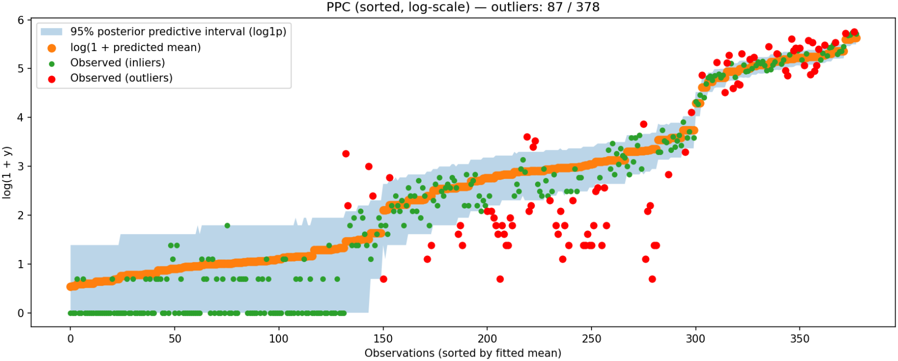




| Intercept | |
|---|---|
| Lower bound | Upper bound |
| -10.851 | -10.291 |



# Posterior Inference and Model Assessment

## Posterior predictive check:



# Hierarchical Generalized Linear Model

**Likelihood:**

$$T_{i,j,k,c} \mid \lambda_{i,j,c}, X_{i,c}, \rho_k, \phi_T \overset{\text{ind}}{\sim} \text{NB}(\rho_k \, \lambda_{i,j,c} \, , \phi_T)$$
$$\lambda_{i,j,c} = \alpha_i \, \theta_{i,j,c}$$

where:

- $\rho_k \, \lambda_{i,j,c}$ = average observed counts
- $\phi_T$ = overdispersion parameter
- $T_{i,j,k,c}$ = transconjugant count
- $X_{i,c}$ = vector containing covariates (1, Temp, [IBU], [DMSO])
- $\rho_k = 10^{-\lceil \frac{k}{3} \rceil}$ dilution factor
- $\alpha_i$ = donor count (exposure) for experiment $i$
- $\theta_{i,j,c}$ = transconjugation rate per donor

Indexes:

- $i \in \{1, ..., 10\}$ = experiment
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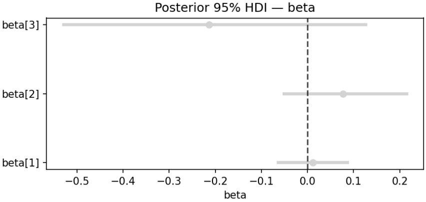
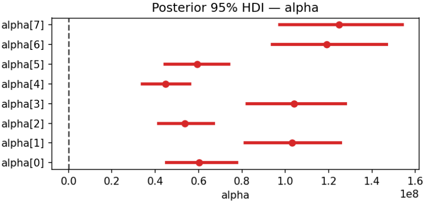
- $k \in \{1, ..., 9\}$ = observations within each experiment replica
- $c \in \{0, 1\}$ = control (0) or treatment (1)

**Overdispersion:**

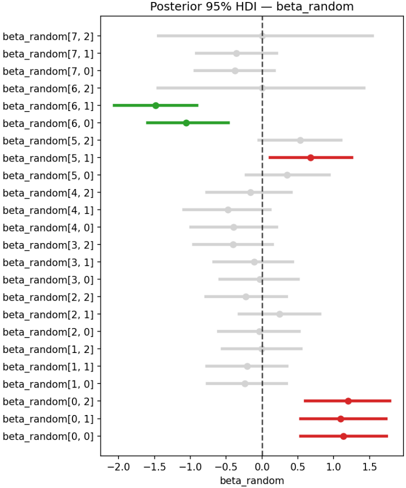
$$\boxed{\phi_T \sim \text{Exp}(0.1)}$$

## Simple Negative Binomial:



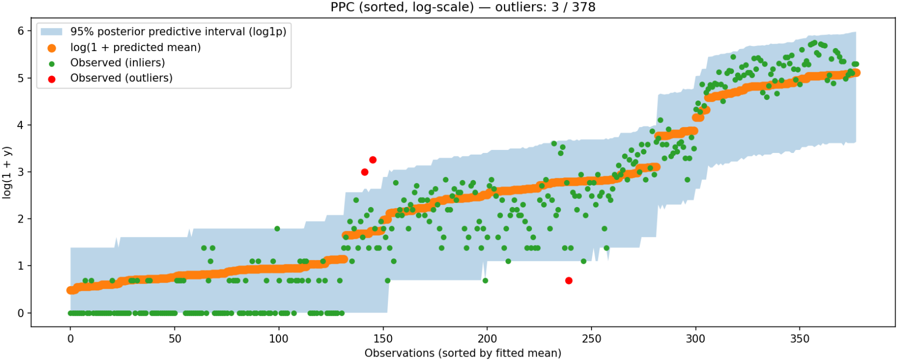




| Intercept | |
|---|---|
| Lower bound | Upper bound |
| -11.162 | -10.488 |



# Posterior Inference and Model Assessment

**Posterior predictive check:**



# Extended model specification

- The linear predictor using the **baseline design matrix X** is

$$b_{i,j} + \beta_0 + \beta_1\,[IBU] + \beta_2\,[DMSO] + \beta_3\,Temp$$

- Adding the **interaction between covariates** $X_{\text{interactions}}$ the linear predictor becomes

$$\begin{aligned} & b_{i,j} + \beta_0 + \beta_1\,[IBU] + \beta_2\,[DMSO] + \beta_3\,Temp \\ & \quad + \beta_4\,[IBU] \times [DMSO] + \beta_5\,[IBU] \times Temp + \beta_6\,[DMSO] \times Temp \end{aligned}$$

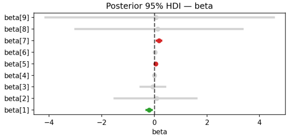
- Adding the **quadratic terms** of the covariates to $X_{\text{interactions}}$, the linear predictor $X_{\text{full}}$ becomes

$$\begin{aligned} & b_{i,j} + \beta_0 + \beta_1\,[IBU] + \beta_2\,[DMSO] + \beta_3\,Temp \\ & \quad + \beta_4\,[IBU] \times [DMSO] + \beta_5\,[IBU] \times Temp + \beta_6\,[DMSO] \times Temp \\ & \quad + \beta_7\,[IBU]^2 + \beta_8\,[DMSO]^2 + \beta_9\,Temp^2 \end{aligned}$$

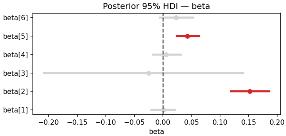
# Shrinkage Model Comparison - Poisson Model

| $\beta$ Prior | X | Intercept | Selected Covariates | Credible Interval |
|---|---|---|---|---|
| Simple | $X_{\text{interactions}}$ | -10.568 $\pm$ 0.273 | DMSO<br>IBU$\times$Temp | 0.155 $\pm$ 0.036<br>0.044 $\pm$ 0.019 |
| Lasso | $X_{\text{interactions}}$ | -10.577 $\pm$ 0.264 | DMSO<br>IBU$\times$Temp | 0.155 $\pm$ 0.035<br>0.044 $\pm$ 0.019 |
| Lasso | $X_{\text{full}}$ | -10.572 $\pm$ 0.273 | IBU<br>IBU$\times$Temp<br>$\text{IBU}^2$ | -0.203 $\pm$ 0.105<br>0.046 $\pm$ 0.019<br>0.177 $\pm$ 0.088 |
| Horseshoe | $X_{\text{interactions}}$ | -10.584 $\pm$ 0.259 | DMSO<br>IBU$\times$Temp | 0.152 $\pm$ 0.034<br>0.043 $\pm$ 0.019 |
| Reg. Horseshoe | $X_{\text{interactions}}$ | -10.581 $\pm$ 0.269 | DMSO<br>IBU$\times$Temp | 0.152 $\pm$ 0.033<br>0.042 $\pm$ 0.020 |
| Reg. Horseshoe | $X_{\text{full}}$ | -10.580 $\pm$ 0.270 | IBU$\times$Temp | 0.043 $\pm$ 0.020 |
| R2D2 | $X_{\text{interactions}}$ | -10.578 $\pm$ 0.272 | DMSO<br>IBU$\times$Temp | 0.154 $\pm$ 0.034<br>0.044 $\pm$ 0.019 |
| Spike&Slab | $X_{\text{interactions}}$ | -10.539 $\pm$ 0.265 | DMSO<br>IBU$\times$Temp | 0.157 $\pm$ 0.031<br>0.032 $\pm$ 0.024 |

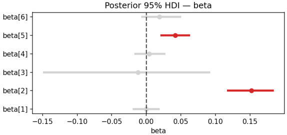
# Shrinkage Model Comparison



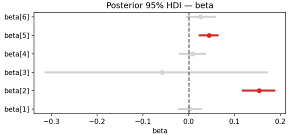

Lasso



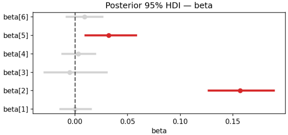

Horseshoe



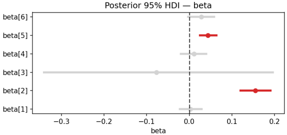

Regularized Horseshoe



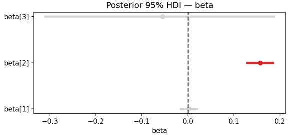

R2D2



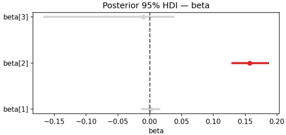

Spike&Slab



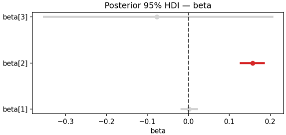

Simple




R2D2 (X)




Spike&Slab (X)




Simple (X)

# Conclusions

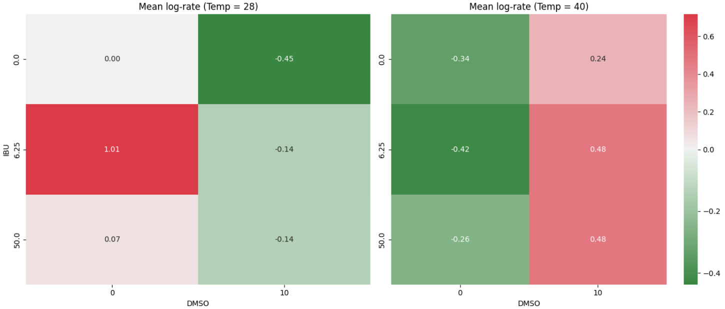



Figure: posterior log-mean effect of combination of IBU and DMSO when Temp is low and high

**Key biological findings:**

- **DMSO** ⇒ negative effect on trasconjugation rate
- **IBU**×**Temp** ⇒ negative effect on trasconjugation rate
- The effect of stressors changes across **temperature levels**

# Conclusions

**Key modeling insights:**

- **Replicate-level variability** is significant $\Rightarrow$ hierarchical random effects are necessary
- **Hierarchical modeling of donor counts** propagates uncertainty from dilution and counting procedures
- **Interaction terms** improve the description of the response surface beyond additive effects
- **Shrinkage priors** identify the most relevant covariates on the response surface

# Thank You!

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# Exploratory analysis

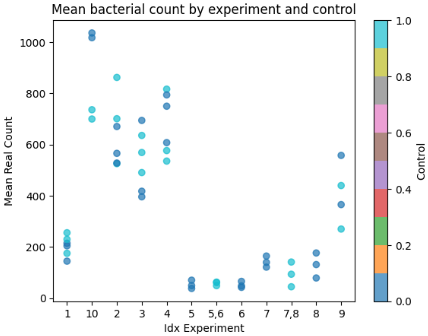


Figure: Mean real count by experiment and control condition.

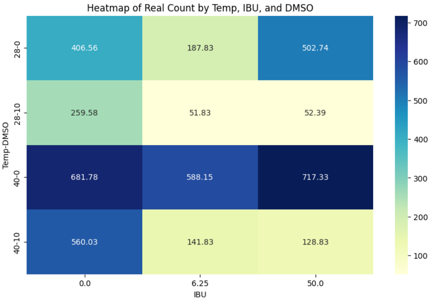


Figure: Mean real count by Temp, DMSO and IBU.

# Shrinkage Model Comparison

| Model | $\beta$ Prior | X Input | Intercept | Selected Covariates |
|---|---|---|---|---|
| Poisson | Simple | X | -10.571 $\pm$ 0.280 | DMSO (0.157 $\pm$ 0.028) |
| Poisson | Simple | $X_{\text{interactions}}$ | -10.568 $\pm$ 0.273 | DMSO (0.155 $\pm$ 0.036), IBU$\times$Temp (0.044 $\pm$ 0.019) |
| Poisson | Lasso | X | -10.578 $\pm$ 0.272 | DMSO (0.157 $\pm$ 0.028) |
| Poisson | Lasso | $X_{\text{interactions}}$ | -10.577 $\pm$ 0.264 | DMSO (0.155 $\pm$ 0.035), IBU$\times$Temp (0.044 $\pm$ 0.019) |
| Poisson | Lasso | $X_{\text{full}}$ | -10.572 $\pm$ 0.273 | IBU (-0.203 $\pm$ 0.105), IBU$\times$Temp (0.046 $\pm$ 0.019), IBU2 (0.1 $\pm$ 0.088) |
| Poisson | Horseshoe | $X_{\text{interactions}}$ | -10.584 $\pm$ 0.259 | DMSO (0.152 $\pm$ 0.034), IBU$\times$Temp (0.043 $\pm$ 0.019) |
| Poisson | Reg. Horseshoe | X | -10.580 $\pm$ 0.278 | DMSO (0.155 $\pm$ 0.028) |
| Poisson | Reg. Horseshoe | $X_{\text{interactions}}$ | -10.581 $\pm$ 0.269 | DMSO (0.152 $\pm$ 0.033), IBU$\times$Temp (0.042 $\pm$ 0.020) |
| Poisson | Reg. Horseshoe | $X_{\text{full}}$ | -10.580 $\pm$ 0.270 | IBU$\times$Temp (0.043 $\pm$ 0.020) |
| Poisson | R2D2 | X | -10.577 $\pm$ 0.275 | DMSO (0.157 $\pm$ 0.028) |
| Poisson | R2D2 | $X_{\text{interactions}}$ | -10.578 $\pm$ 0.272 | DMSO (0.154 $\pm$ 0.034), IBU$\times$Temp (0.044 $\pm$ 0.019) |
| Poisson | Spike&Slab | X | -10.537 $\pm$ 0.269 | DMSO (0.157 $\pm$ 0.028) |
| Poisson | Spike&Slab | $X_{\text{interactions}}$ | -10.539 $\pm$ 0.265 | DMSO (0.157 $\pm$ 0.031), IBU$\times$Temp (0.032 $\pm$ 0.024) |

# Sensitivity Analysis

| Prior type | Prior params | $\mu_{\alpha_0}$ | $\sigma_{\alpha_0}$ | Covariates | LOO |
|---|---|---|---|---|---|
| Inverse-Gamma | shape=3, scale=0.18 | 0 | 5 | $X_{\text{interactions}}$ | -1602.81 |
| Inverse-Gamma | shape=3, scale=0.72 | 0 | 5 | $X_{\text{interactions}}$ | -1602.94 |
| Inverse-Gamma | shape=3, scale=0.18 | 0 | 10 | $X_{\text{interactions}}$ | -1603.13 |
| Half-Student t | df=3, scale=0.5 | 0 | 1 | $X_{\text{interactions}}$ | -1603.23 |
| Half-Normal | $\sigma_b = 0.5$ | 0 | 10 | $X_{\text{interactions}}$ | -1603.32 |
| Inverse-Gamma | shape=3, scale=2.0 | 0 | 10 | $X_{\text{interactions}}$ | -1603.54 |
| Half-Normal | $\sigma_b = 0.5$ | 0 | 1 | $X_{\text{interactions}}$ | -1603.81 |
| Inverse-Gamma | shape=3, scale=0.72 | 0 | 10 | $X_{\text{interactions}}$ | -1603.87 |
| Half-Normal | $\sigma_b = 1$ | 0 | 10 | $X_{\text{interactions}}$ | -1604.00 |
| Half-Student t | df=3, scale=0.5 | 0 | 10 | $X_{\text{interactions}}$ | -1604.24 |
| Inverse-Gamma | shape=3, scale=2.0 | 0 | 5 | $X_{\text{interactions}}$ | -1604.39 |
| Inverse-Gamma | shape=3, scale=0.18 | 0 | 5 | $X$ | -1608.18 |
| Inverse-Gamma | shape=3, scale=2.0 | 0 | 10 | $X$ | -1608.52 |
| Half-Normal | $\sigma_b = 1$ | 0 | 10 | $X$ | -1608.57 |
| Half-Normal | $\sigma_b = 0.5$ | 0 | 1 | $X$ | -1608.61 |
| Half-Normal | $\sigma_b = 0.5$ | 0 | 10 | $X$ | -1608.65 |
| Inverse-Gamma | shape=3, scale=0.18 | 0 | 10 | $X$ | -1608.73 |
| Inverse-Gamma | shape=3, scale=0.72 | 0 | 5 | $X$ | -1608.80 |
| Inverse-Gamma | shape=3, scale=0.72 | 0 | 10 | $X$ | -1608.89 |
| Half-Student t | df=3, scale=0.5 | 0 | 10 | $X$ | -1609.51 |
| Half-Student t | df=3, scale=0.5 | 0 | 1 | $X$ | -1609.56 |
| Inverse-Gamma | shape=3, scale=2.0 | 0 | 5 | $X$ | -1610.25 |

# Divergence Rates

| Base Model | $\beta$ Prior | X Input | Divergence Rate |
|---|---|---|---|
| Poisson | Gaussian | X | 0.00 |
| Poisson | Gaussian | $X_{interactions}$ | 0.00 |
| Poisson | Lasso | X | 0.00 |
| Poisson | Lasso | $X_{interactions}$ | 0.00 |
| Poisson | Lasso | $X_{full}$ | 0.00 |
| Poisson | Horseshoe | $X_{interactions}$ | 1.09 |
| Poisson | Regularized Horseshoe | X | 1.05 |
| Poisson | Regularized Horseshoe | $X_{interactions}$ | 1.95 |
| Poisson | Regularized Horseshoe | $X_{full}$ | 3.43 |
| Poisson | R2D2 | X | 0.45 |
| Poisson | R2D2 | $X_{interactions}$ | 0.61 |
| Poisson | Spike&Slab | X | 0.00 |
| Poisson | Spike&Slab | $X_{interactions}$ | 0.00 |
| Negative Binomial | Gaussian | X | 0.00 |
| Negative Binomial | Gaussian | $X_{interactions}$ | 0.00 |
| Negative Binomial | Lasso | X | 0.00 |
| Negative Binomial | Lasso | $X_{interactions}$ | 0.00 |
| Negative Binomial | Horseshoe | X | 1.50 |
| Negative Binomial | Horseshoe | $X_{interactions}$ | 2.60 |
| Negative Binomial | R2D2 | X | 1.16 |
| Negative Binomial | Spike&Slab | X | 0.00 |
| Negative Binomial | Half-Normal | X | 0.00 |

# Shrinkage Model Comparison - Negative Binomial

| $\beta$ **Prior** | **X Input** | **Intercept** | **Selected Covariates** |
|---|---|---|---|
| Gaussian | X | -10.825 $\pm$ 0.337 | — |
| Gaussian | $X_{interactions}$ | -10.822 $\pm$ 0.306 | — |
| Lasso | X | -10.829 $\pm$ 0.330 | — |
| Lasso | $X_{interactions}$ | -10.829 $\pm$ 0.313 | — |
| Horseshoe | X | -10.839 $\pm$ 0.329 | — |
| Horseshoe | $X_{interactions}$ | -10.838 $\pm$ 0.314 | — |
| R2D2 | X | -10.829 $\pm$ 0.332 | — |
| Spike&Slab | X | -10.764 $\pm$ 0.336 | — |
| Half-Normal | X | -10.826 $\pm$ 0.345 | — |